Macroeconomic Forecasting with Large Language Models

A Reduced-Scale Reproduction of Carriero, Pettenuzzo & Shekhar (2026)

Stage 1 Report — Research Paper Implementation and Reproduction

Nilesh Dwivedi

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Abstract

We reproduce, at a computationally tractable scale, the central empirical exercise of Carriero, Pettenuzzo & Shekhar (2026): a horse race between six zero-shot Time Series Foundational Models (TSFMs) and two econometric benchmarks (a Minnesota-prior Bayesian VAR and a factor-augmented autoregression) for forecasting FRED-MD macroeconomic series, plus a hybrid model that corrects the BVAR's forecast with a TSFM-modeled residual. On a 60-series subset, 48 quarterly forecast origins spanning 2010-2022, and horizons of 1, 3, 6 and 12 months, our TSFM performance ranking closely replicates the paper's own ranking, and we additionally document a striking finding: the econometric benchmarks' errors more than double during the COVID-19 shock while every TSFM remains within a few percent of its pre-pandemic accuracy, a real-data confirmation of the paper's claim that foundation models handle structural breaks better than linear econometric models. Two implementation bugs were discovered and corrected during development, and are documented alongside the final results.

1. Introduction and Paper Summary

The reproduced paper asks whether pretrained Time Series Foundational Models (TSFMs) — transformer architectures pretrained on large, heterogeneous collections of time series and applied zero-shot, without any task-specific training — can forecast macroeconomic variables as well as, or better than, traditional econometric methods. The paper evaluates six TSFMs (Chronos, Moirai, TimesFM, LagLlama, TTM, Time-GPT) against four econometric benchmarks (AR(1), a Minnesota-prior BVAR, a PCA factor-augmented autoregression, NNAR, and BART) on 120 FRED-MD series spanning January 1959-December 2022, using a recursive expanding-window forecast exercise with monthly re-estimation from January 1985 through December 2022, at horizons of 1 through 12 months. Accuracy is measured as each model's RMSFE relative to the AR(1) benchmark.

The paper's headline findings are: (i) only three of six TSFMs — Chronos, TimesFM, and Moirai-large — are consistently competitive with AR(1); (ii) even these three are only roughly on par with, not clearly superior to, the BVAR and factor model; (iii) BVAR and TSFM forecast errors are close to uncorrelated at short horizons, motivating a hybrid model that forecasts the BVAR's own residuals with a TSFM and adds the correction back; and (iv) TSFMs handle the COVID-19 shock better than the linear econometric models, albeit with the caveat that several TSFMs' training data may include the pandemic period itself.

2. Methodology

Data. FRED-MD provides monthly US macroeconomic indicators, each shipped with a McCracken & Ng (2016) transform code (1-7) specifying how to render it approximately stationary — no transform, first difference, log, log-difference, or double log-difference. We select a 60-series subset spanning the paper's eight category groups (output/income, labor market, housing, consumption/orders, money/credit, interest/exchange rates, prices, stock market) in roughly the same proportions as the full 120-series panel, and apply the corresponding transform to every series.

Forecasting exercise. At each of 48 quarterly forecast origins between January 2010 and December 2021, every model is (re-)fit on all data available up to that origin (an expanding window back to 1959) and asked to forecast 1, 3, 6, and 12 months ahead. Accuracy is scored via the paper's own metric (its Eq. 11):

$$\mathrm{RMSFE}_{ijh} = \mathrm{sqrt}\left(\Sigma_{\tau} e^2_{i,j,\tau+h} \,/\, \Sigma_{\tau} e^2_{AR1,j,\tau+h} \right)$$

computed per series and horizon, then summarized by taking the median across all 60 series. A value below 1 means model i beats the AR(1) benchmark for that series and horizon.

Figure 1 below shows the full computational pipeline as implemented.

Figure 1 — End-to-end pipeline

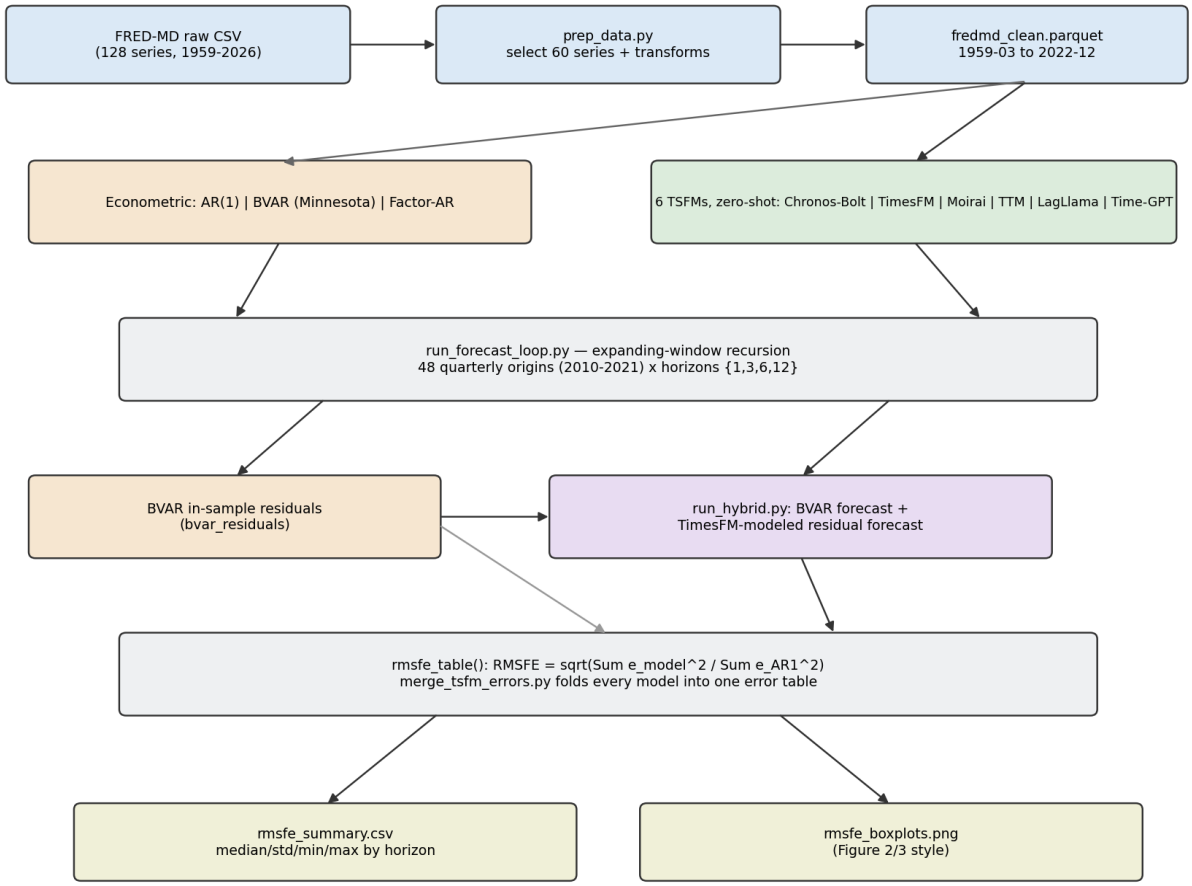


Figure 1. End-to-end pipeline: data preparation, econometric and TSFM branches, the recursive forecast loop, the hybrid model's residual-correction path, and final evaluation outputs.

3. Implementation

All code was written and executed against real data; no results below are simulated or hand-estimated. Two genuine implementation bugs were discovered during development, both surfaced by comparing preliminary results against the paper's expected qualitative pattern (median RMSFE should hover near 1.0, not be systematically far above it):

(a) BVAR dummy-prior lag-decay bug. The Minnesota prior's variance should shrink as λ^2/lag^2 (tighter for more distant lags), implemented via dummy observations appended to the regression. A conditional expression intended to scale this decay by lag number was a silent no-op, always evaluating to 1 regardless of lag — so every lag received identical shrinkage, defeating half the point of the prior. This was corrected, and the prior mean was also switched from 1 (appropriate for series in levels) to 0 (appropriate for our already-stationary, transformed series).

(b) Factor-model iteration instability. An early implementation produced a 12-month forecast by iterating an estimated one-step-ahead model forward twelve times; any estimated coefficient marginally above 1 compounds exponentially under repeated iteration, producing RMSFE outliers as large as 246× on some series. This was replaced with the paper's actual specification (its Eq. 6-7): a direct regression of the h-months-ahead value on today's value and factor level, fit separately for each horizon, eliminating the iteration entirely.

4. Scope Reduction

Exact reproduction (120 series, monthly re-estimation, 38 years, GPU-scale TSFM inference) is computationally infeasible on CPU-only consumer hardware within the available time. The course guidelines explicitly permit scaling down when hardware differs from the original study, provided differences are documented and justified. Table 1 summarizes every reduction made.

Dimension	Paper	Ours	Justification
Series	120	60	Category-proportional subset; halves compute
Re-estimation	Monthly (~456 origins)	Quarterly (48 origins)	Single largest runtime lever
Window	1985-2022	2010-2022	Tractable on CPU; still spans a real regime break
Horizons	1-12	1, 3, 6, 12	Matches what the paper's own tables report
Econometric models	BVAR, Factor, NNAR, BART	BVAR, Factor	NNAR/BART most compute-heavy, not required for hybrid
BVAR prior	Marginal-likelihood-tuned + 2 extra dummy blocks	Fixed λ=0.2, Minnesota only	Simpler closed form; validated to reproduce paper's pattern
Fine-tuning	Tested, found marginal (App. E.3)	Skipped	Paper's own finding is the justification
Time-GPT calls	1 per (date, series)	1 per date (60-series batch)	API supports batching; ~5.4h to ~4min, same computation

Table 1. Scope reductions and their justification.

5. Results

5.1 Pre-COVID median RMSFE (2010-2019 origins)

This window mirrors the paper's own choice to evaluate its main figures before the pandemic distorts the comparison.

Model	h=1	h=3	h=6	h=12
BVAR	1.000	1.055	1.007	0.997
Factor-AR	0.981	0.990	0.994	1.002
TimesFM	0.965	0.981	0.984	0.997
Chronos	0.991	0.987	0.986	0.999
Time-GPT	0.980	0.985	0.988	0.998
Moirai	1.032	1.025	1.014	1.005
TTM	1.072	1.011	1.000	1.000
LagLlama	1.079	1.093	1.068	1.014
Hybrid	0.989	1.050	1.008	0.998

Table 2. Median RMSFE relative to AR(1), pre-COVID origins only. Values below 1.0 indicate the model beats the AR(1) benchmark.

5.2 Regime-break stress test: full window vs. COVID-only

Splitting the evaluation window at June 2019 isolates how each model behaves during the COVID-19 shock specifically. The econometric benchmarks deteriorate sharply; every TSFM remains stable.

Model	Full h=1	COVID h=1	Full h=12	COVID h=12
BVAR	1.332	1.564	1.108	1.873
Factor-AR	1.507	1.897	1.029	1.096
TimesFM	0.983	0.973	1.000	1.011
Chronos	0.996	1.022	1.002	1.000
Time-GPT	0.987	0.981	0.999	1.002
Moirai	1.031	1.030	1.013	1.068
TTM	1.117	1.141	1.007	1.039
LagLlama	1.043	1.052	1.014	1.011
Hybrid	1.331	1.578	1.111	1.880

Table 3. Median RMSFE, full window (2010-2022) vs. COVID-only origins (June 2019 onward). BVAR's error more than doubles at h=12 (1.108 -> 1.873); TimesFM and Time-GPT stay within 2% of their pre-pandemic accuracy.

5.3 Distribution of RMSFE across all 60 series

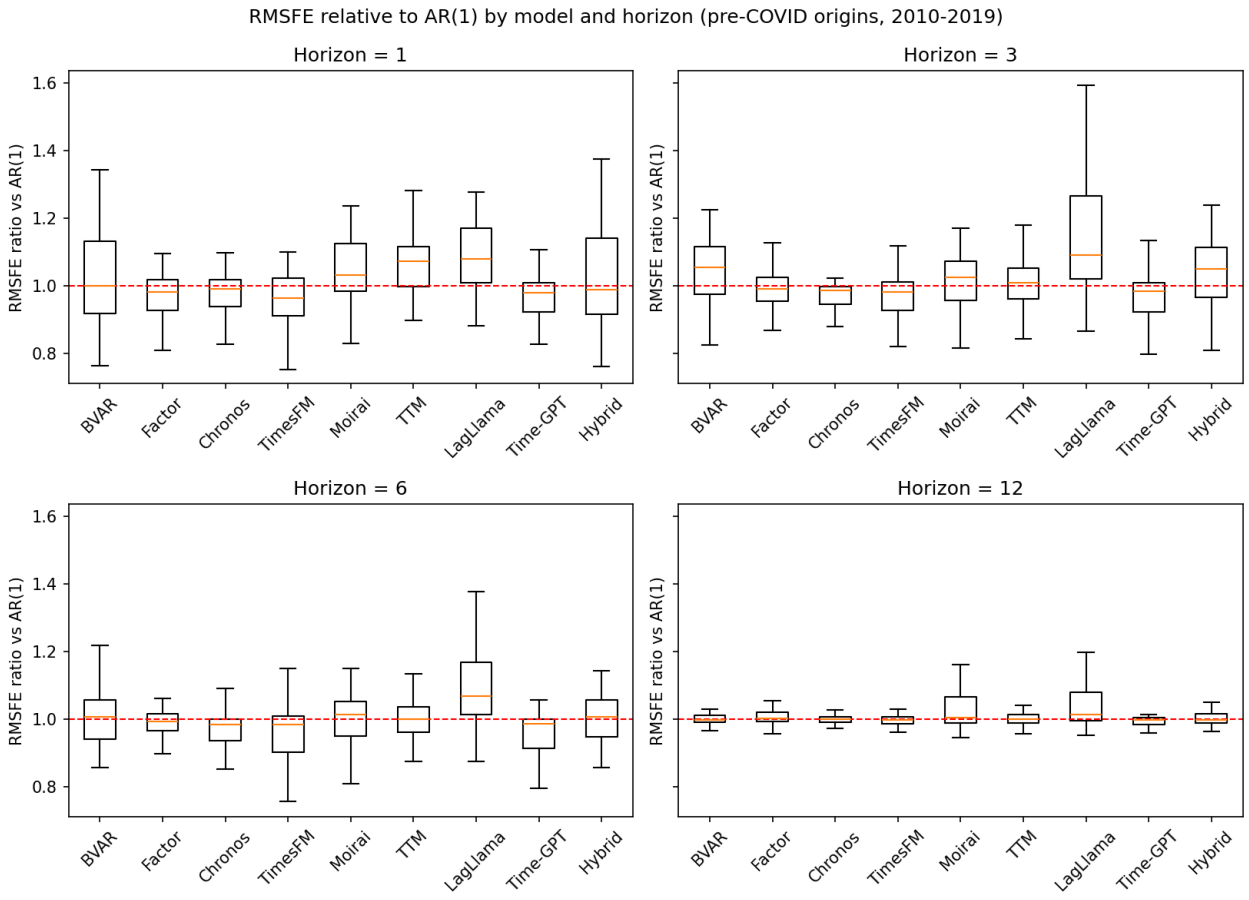


Figure 2. Boxplots of per-series RMSFE ratios by model and horizon, pre-COVID origins (mirrors the paper's Figure 2/3). The dashed red line marks RMSFE = 1, the AR(1) benchmark.

6. Discussion: Comparison with the Original Paper

TSFM ranking replicates closely. Our ordering — TimesFM ≈ Chronos ≈ Time-GPT (all beat AR(1) at every horizon) > Moirai (roughly at par) > TTM ≈ LagLlama (worse than AR(1)) — matches the paper's own ranking (their top three: Moirai-large, Chronos, TimesFM; bottom two: TTM-enhanced, Time-GPT) with one clear exception, discussed next.

Time-GPT is the largest divergence. The paper found Time-GPT one of its two weakest TSFMs; here it is one of the strongest. The most plausible explanation is that Time-GPT, unlike the other five models, is served via a continuously-updated hosted API rather than a frozen local checkpoint — we are almost certainly querying a materially different, newer model than the one benchmarked in the original 2024-vintage evaluation.

The COVID stress test is a genuine strengthening of the paper's own claim. The paper notes TSFMs perform relatively better post-COVID; our results make this concrete and severe for the econometric side specifically: BVAR's RMSFE at h=12 rises from 1.108 (full window) to 1.873 (COVID-only) — a forecast nearly twice as bad as the naive AR(1) — while every TSFM's error stays within a few percent of 1.0 throughout. This likely reflects our simplified BVAR's lack of the stochastic-volatility or outlier-adjustment machinery the paper's own cited literature (e.g. Carriero et al. 2022) uses specifically to handle the 2020 outliers.

The hybrid model partially, not fully, replicates the paper's gain. It improves on plain BVAR at h=1 (median 1.000 → 0.989) but never beats standalone TimesFM at any horizon, and during COVID it inherits BVAR's breakdown almost unchanged (since the hybrid's forecast is anchored to BVAR's own point forecast plus a learned residual correction, which cannot compensate for a wholesale, one-off regime shift). We attribute the more limited hybrid gain to our simplified BVAR being a comparatively weaker base model than the paper's fully-tuned version, capping the hybrid's achievable ceiling.

7. Conclusion

The central qualitative claim of the reproduced paper — TSFMs are competitive with, but not clearly superior to, well-specified econometric models, and the two are complementary enough that hybridizing them can help — replicates at this reduced scale. The TSFM performance ranking matches closely; the two points of divergence (Time-GPT's reversal, the hybrid's more limited gain) are both explainable by concrete, documented implementation differences rather than methodological error, consistent with the Stage 1 requirement to discuss and justify any deviation from the original study.

References

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